UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

PHYSICS – IMPORTANT FORMULAE & CONSTANTS

GRADE 12 ADVANCED | TERM 1 | 2026–2027

MODULE 10 | FORMS OF ENERGY & MECHANICAL ENERGY

TRANSLATIONAL KINETIC ENERGY

$$KE = \frac{1}{2}mv^2$$

KE = kinetic energy (J); m = mass (kg); v = speed (m/s).

ROTATIONAL KINETIC ENERGY

$$KE_{\text{rot}} = \frac{1}{2}I\omega^2$$

I = moment of inertia (kg·m²); ω = angular speed (rad/s); KE_{rot} in joules (J).

GRAVITATIONAL POTENTIAL ENERGY

$$GPE = mgh$$

m = mass (kg); g = 9.8 m/s²; h = height above a chosen reference level (m).

CHANGE IN GPE

$$\Delta GPE = mg(h_f - h_i)$$

ΔGPE = final minus initial gravitational potential energy (J). Use the same reference level.

WORK DONE BY GRAVITY

$$W_g = -\Delta GPE = mg(h_i - h_f)$$

W_g in joules (J). Falling → gravity does positive work; rising → negative work.

MECHANICAL ENERGY

$$ME = KE + PE$$

Add all kinetic and potential-energy terms included in the chosen model (J).

CONSERVATION OF MECHANICAL ENERGY

$$KE_i + PE_i = KE_f + PE_f$$

Use when friction and other energy transfers out of mechanical energy are negligible.

REST ENERGY

$$E_0 = mc^2$$

E_0 = rest energy (J); m = mass (kg); c = 3.00 × 10^8 m/s. Mass is a form of energy.

MODULE 10 | USE THE CORRECT ENERGY MODEL

SITUATION	REMEMBER
Rolling without slipping	Total KE = ½mv² + ½Iω²; ω = v/R. R = rolling radius (m).
Friction / deformation	Mechanical energy can decrease while total energy remains conserved. Track thermal/internal energy.

MODULE 10 | ENERGY CONSERVATION — USEFUL FORMS

GRAVITY + TRANSLATIONAL MOTION

$$\frac{1}{2}mv_i^2 + mgh_i = \frac{1}{2}mv_f^2 + mgh_f$$

Frictionless motion; ignore rotation. The reference height is arbitrary but must be consistent.

SPEED AFTER A VERTICAL DROP

$$v_f^2 = v_i^2 + 2g(h_i - h_f)$$

g = positive magnitude, 9.8 m/s². Use when the only energy exchange is between KE and GPE.

HEIGHT GAIN WHEN COMING TO REST

$$h_f - h_i = \frac{v_i^2}{2g}$$

For an object stopping after rising, without friction and without rotational energy.

WORK-ENERGY RELATION

$$W_{\text{net}} = KE_f - KE_i$$

W_{net} = net work (J). Supporting tool for energy calculations.

CONSTANT-FORCE WORK

$$W = F\Delta x\cos\theta$$

F = force (N); Δx = displacement magnitude (m); θ = angle between force and displacement.

ELASTIC POTENTIAL ENERGY

$$EPE = \frac{1}{2}kx^2$$

For an ideal Hooke's-law spring: k = spring constant (N/m); x = stretch or compression (m).

MODULE 10 | ROLLING ENERGY & MOMENTS OF INERTIA

ROLLING WITHOUT SLIPPING

$$KE = \frac{1}{2}mv^2 + \frac{1}{2}I\frac{v^2}{R^2}$$

Use ω = v/R. For a solid sphere, I = ⅖mR², so KE = ⁷⁄₁₀mv².

PENDULUM AFTER AN IMPACT

$$\frac{1}{2}Mv^2 = Mgh \quad v = \sqrt{2gh}$$

During a frictionless swing after impact: M = combined mass (kg), h = rise (m).

OBJECT / AXIS	MOMENT OF INERTIA
Hoop about its symmetry axis	I = mR²
Solid uniform disk / cylinder, symmetry axis	I = ½mR²
Solid uniform sphere, central axis	I = ⅖mR²

Boundary: energy conservation can connect the states before and after a frictionless motion. In an inelastic impact itself, use momentum first; kinetic energy is not conserved.

MODULE 10 | COLLISIONS & MOMENTUM

LINEAR MOMENTUM

$$p = mv$$

p = signed momentum (kg·m/s); m = mass (kg); v = velocity (m/s).

CONSERVATION OF MOMENTUM

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Applies if net external impulse is negligible. u = initial velocity; v = final velocity (m/s).

PERFECTLY INELASTIC COLLISION

$$v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$$

Objects stick together and share v. This follows from momentum conservation.

INITIAL TOTAL KINETIC ENERGY

$$KE_i = \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2$$

Add the kinetic energies of both objects (J). Squared speeds make KE non-negative.

FINAL KE — JOINED OBJECTS

$$KE_f = \frac{1}{2}(m_1 + m_2)v^2$$

Combined mass (kg); shared speed (m/s); final KE in joules (J).

KINETIC ENERGY TRANSFORMED

$$E_{\text{transformed}} = KE_i - KE_f$$

For an inelastic impact, this decrease appears mainly as internal/thermal energy, deformation and sound.

MODULE 10 | CHOOSE THE CONSERVATION LAW

STAGE / TYPE	MOMENTUM	KINETIC / MECHANICAL ENERGY
Isolated elastic impact	Conserved	Total KE conserved.
Isolated inelastic impact	Conserved	Total KE is not conserved.
Perfectly inelastic impact	Conserved	Objects stick; KE decreases.
Frictionless upward swing	Gravity gives external impulse	ME conserved during the swing.

Impact followed by a swing: (1) find the post-impact speed using the rise: $v = \sqrt{2gh}$; (2) apply momentum conservation to the short impact. Choose one positive direction for every velocity.

QUICK REFERENCE | MECHANICS UNITS

QUANTITY	SI UNIT
Energy and work	joule: J = kg·m²/s² = N·m
Momentum / force	kg·m/s / newton (N)
Mass / height / speed	kg / m / m/s

MODULE 22 | PHOTONS, QUANTIZATION & PHOTOELECTRIC EFFECT

WAVE SPEED IN VACUUM

$$c = f\lambda$$

c = 3.00 × 10^8 m/s; f = frequency (Hz); λ = vacuum wavelength (m).

PHOTON ENERGY

$$E_\gamma = hf = \frac{hc}{\lambda}$$

E_γ = photon energy (J); h = Planck constant (J·s). Shorter λ → greater photon energy.

QUANTIZED OSCILLATOR ENERGY

$$E_n = nhf$$

n = 0, 1, 2, …; E_n = oscillator energy (J). A one-step energy change is hf.

CONVENIENT eV–nm FORM

$$E_\gamma(\text{eV}) = \frac{1240}{\lambda(\text{nm})}$$

Use this form only with λ in nm and E in eV. hc ≈ 1240 eV·nm.

PHOTOELECTRIC ENERGY BALANCE

$$KE_{\max} = hf - \phi$$

KE_{max} = maximum photoelectron KE; φ = metal work function. Keep energy units consistent.

THRESHOLD FREQUENCY

$$\phi = hf_0 \qquad f_0 = \frac{\phi}{h}$$

f_0 = minimum emission frequency (Hz). No emission when $f < f_0$, regardless of intensity.

THRESHOLD WAVELENGTH

$$\lambda_0 = \frac{hc}{\phi}$$

λ_0 = longest wavelength that can eject electrons (m). Emission possible for $\lambda \le \lambda_0$.

STOPPING POTENTIAL

$$KE_{\max} = eV_s$$

e = elementary-charge magnitude (C); V_s = stopping potential (V). KE is in J here.

MODULE 22 | PHOTOELECTRIC GRAPH & ENERGY UNITS

RELATION / RULE	READ IT AS
$KE_{max} = h(f - f_0)$	Graph of KE in J versus f in Hz: slope = h, x-intercept = f_0, y-intercept of extended line = −φ.
Electron volt	1 eV = 1.602 × 10^{-19} J; J → eV: divide by 1.602 × 10^{-19}.
Intensity at fixed $f > f_0$	More photoelectrons per second; KE_{max} and stopping potential stay unchanged.

At threshold: KE_{max} = 0. Below threshold, the negative result of hf − φ means **no emission**, not negative kinetic energy.

MODULE 22 | MOMENTUM, MATTER WAVES & ATOMIC ENERGY

PHOTON MOMENTUM

$$p_\gamma = \frac{h}{\lambda} = \frac{E_\gamma}{c}$$

p_γ = photon momentum (kg·m/s). Use λ in m and E_γ in J.

DE BROGLIE WAVELENGTH

$$\lambda = \frac{h}{p} \simeq \frac{h}{mv}$$

Matter wavelength (m); p = momentum (kg·m/s). p ≈ mv only for nonrelativistic particles.

HYDROGEN ENERGY LEVEL

$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

n = 1, 2, 3, … . E = 0 at ionization; a bound-state energy is negative.

HYDROGEN ORBITAL RADIUS

$$r_n = n^2 r_1 \quad r_1 = 0.053\ \text{nm}$$

r_n = radius (m or nm). Radius grows as n^2; this is the hydrogen Bohr model.

ATOM'S ENERGY CHANGE

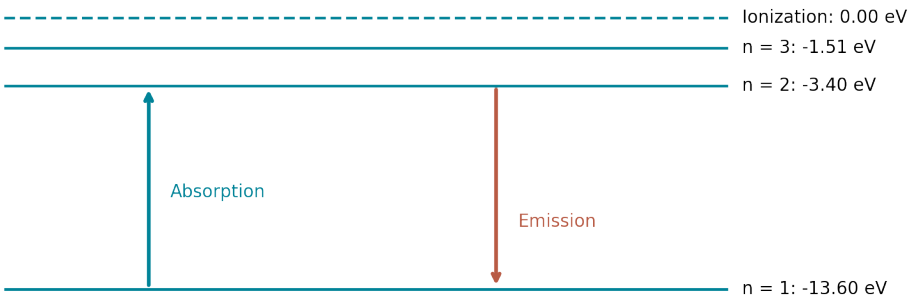
$$\Delta E_{\text{atom}} = E_f - E_i$$

Absorption: ΔE > 0. Emission: ΔE < 0. Use the energy difference, not a single level's energy.

TRANSITION PHOTON

$$E_\gamma = |E_f - E_i| = hf = \frac{hc}{\lambda}$$

Photon energy is positive. For emission, $E_\gamma = E_i - E_f$



QUICK REFERENCE | CONSTANTS & SPECTRAL SERIES

CONSTANT / SERIES	VALUE / DESTINATION
h / c	6.63×10^{-34} J·s ≈ 4.14×10^{-15} eV·s / 3.00×10^{8} m/s
e / m_e	1.602×10^{-19} C / 9.11×10^{-31} kg
Lyman / Balmer / Paschen	Transitions ending at n = 1 / 2 / 3: ultraviolet / mainly visible / infrared.
Unit conversions	1 nm = 10^{-9} m; 1 eV = 1.602×10^{-19} J.

Concept check: greater precision in position means greater uncertainty in momentum. In Compton scattering, energy and momentum are conserved. Higher temperature shifts thermal radiation toward higher peak frequency.